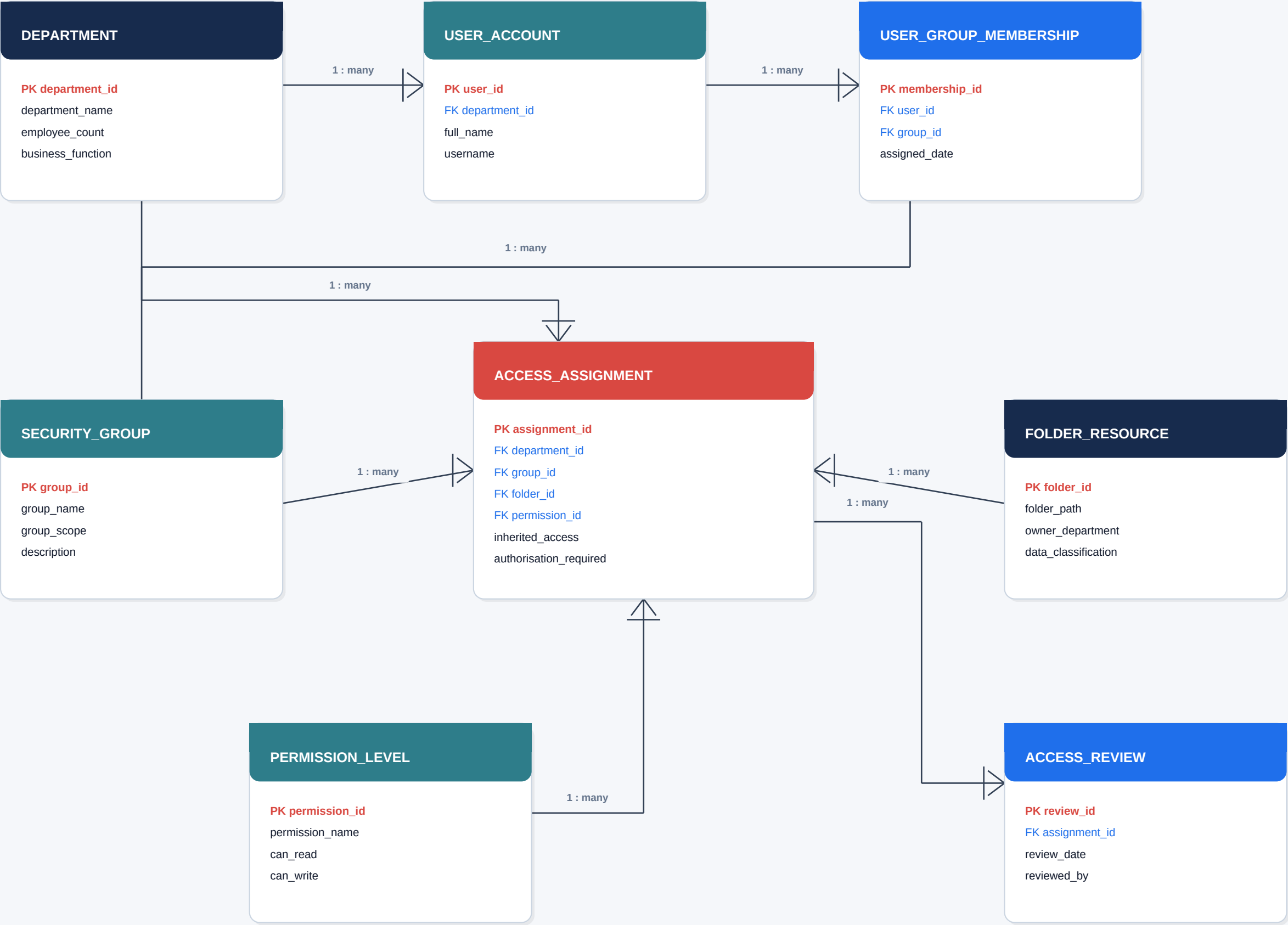


Task 1 - User Permissions Entity Relationship Diagram

Ubuntu Innovations (Pty) Ltd | Tech Career Accelerator Week 2 Deliverable

ERD VERSION - FIXED LAYOUT



Legend

PK = Primary Key
FK = Foreign Key
1 : many = one record can relate to many records in the connected table.

Main junction entity:
ACCESS_ASSIGNMENT

Purpose:
Links departments, security groups, folder resources and permission levels into one auditable access rule.

ERD Notes and Data Mapping

This page explains how the original user permissions matrix was translated into an Entity Relationship Diagram.

Original Matrix Element	ERD Representation	Reason
Department	DEPARTMENT	Stores business areas such as Management, Finance, HR, Sales and Development.
Security Group	SECURITY_GROUP	Stores access groups such as mgmt_grp, finance_grp and all_staff.
Folder Access	FOLDER_RESOURCE	Stores shared folders such as /Company/Finance and /Company/Public.
Permission Level	PERMISSION_LEVEL	Stores reusable permission definitions such as Full Control, Read / Write and Read Only.
Department + Group + Folder + Permission	ACCESS_ASSIGNMENT	Junction entity that records which group receives which permission to which folder.
User account provisioning	USER_ACCOUNT + USER_GROUP_MEMBERSHIP	Supports assigning users to departments and groups using least-privilege access.
Quarterly access review	ACCESS_REVIEW	Tracks periodic verification of group membership and access rights.

Example Access Assignment Records

Department	Group	Folder	Permission
Management	mgmt_grp	/Company/Management (+ inherited folders)	Full Control
Finance	finance_grp	/Company/Finance	Read / Write
HR	hr_grp	/Company/HR	Read / Write
Sales	sales_grp	/Company/Sales	Read / Write
Development	dev_grp	/Company/Development	Read / Write
All Staff	all_staff	/Company/Public	Read Only

Assumptions

- A user belongs to one department but may be assigned to more than one security group when authorised.
- The ACCESS_ASSIGNMENT entity prevents duplication by storing group-folder-permission rules in one place.
- The ACCESS_REVIEW entity supports quarterly access reviews and creates an audit trail.
- Service and system accounts can be modelled separately if the project scope expands.